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2 tangible user interface integration in engineering

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4 **Abstract** In this paper we describe the design and the development of two specific tangible user interface (TUI) platforms. The aim of the first one is to support computer aided design (CAD) parts assembly operations in the mechanical product domain. The aim of the second one is mainly designed to help stakeholders during the task of validation of subsoil model in the field of geosciences. In this paper, we propose a design methodology of the tangible parts (also called props) based on our previous experiences in TUI development. This methodology is mainly based on multidisciplinary work and user tests and we assume that this process is reproducible.

5 **Keywords** Tangible user interface (TUI) · Experiences feedback · Multidisciplinary design methodology · CAD parts assembly · Geosciences

6 Introduction

7 Computer Human Interaction systems have recently appear proposing new concepts for hardware as well as for software. Among these new technologies, the tangible user interface (called TUI in following) seeks to make intuitive interfaces in order to combine the physical world and the digital one to simplify interaction. TUI systems are based on the use of real and physical objects which allow a representation of the data and/or a physical control of digital information. In 1997, Ishii and Ullmer [7] gave a first vision of tangible inter-

11 on: “To make computing truly ubiquitous and invisible, we seek to establish a new type of HCI that we call “Tangible User Interfaces”. TUIs will augment the real physical world by coupling digital information to everyday physical objects and environments”. In 2000 in [14] they characterized more particularly the tangible parts of the TUI, also called props, as a way to “give physical form to digital information, employing physical artifacts both as representations and controls for computational media”. In 2002 Aliakseyeu et al. [1] and then Subramanian in 2004 [13] proposed props specific to navigation and manipulation of 3D data which are the general context of the TUI design presented in this article.

12 The research question we tackled is the design and development of TUI dedicated to industrial domains in order to perform one or more complex tasks.

13 Our standpoint is that in specific and industrial domains, any TUI has to be designed in collaboration with people who have the knowledge and know-how. On this point we can cite [3] “A real-world application domain has specific properties, constraints, and requirement that should be taken into account when designing interaction techniques for that domain”. Thus, the developments of TUI systems adapted to specific industrial tasks are often complex as it implies different skills to design all the system components.

14 This paper focuses on the design phase of the development of TUI systems. Indeed, the development of a TUI system implies the design of different components:

- 15 The tangible parts (called props in this paper) which are the physical components of the system. These props are the entities that will be manipulated by the users.
- The software and hardware systems that will be used to track the positions and movements of props.
 - The interface system that provides to the user a feedback of the props operations in its environment.

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